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Roll No

MI-7002-CBGS

B.E. VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Strata Control

Time : Three Hours

Maximum Marks : 70

Note : i) Attempt any five questions.

ii) All questions carry equal marks.

1. How roofs are examined? Explain roof stitching and roof bolting?
2.
 - a) Describe a powered support and its principles of operation in details.
 - b) Describe the constructional features and application of the powered support.
3.
 - a) Describe a hydraulic prop and its working principle. Compare it with a friction prop.
 - b) How roof is tested in a coal mine? What are the indications of a “bad” roof?
4. What are the advantages of stowing? Explain the different types of stowing in details (with sketch).
5.
 - a) Write a note on the behaviour of rock pressure during narrow and wide excavation.
 - b) Explain the theories of “Ground movement”.

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6. Discuss, Rock, Burst, Bumps, Gas outbursts and pot holes.
7.
 - a) Explain a neat sketch indicating caving, fracturing and subsidence in the strata over a Longwall panel and explain in detail the developments taking place therein.
 - b) Explain the impacts of subsidence in details.
8. Write notes on (any three)
 - a) Strata Control and Monitoring Plan
 - b) Support plan
 - c) Withdrawal of support
 - d) Pipe jam in hydraulic stowing
